

## Assignment 1: Machine learning

### Question 1: Introduction to Machine Learning

a) Define Machine Learning in your own words.

Ans: It is a branch of AI in which computers learn patterns from data and make predictions or decisions without being explicitly programmed for every task.

b) Differentiate between supervised and unsupervised learning with suitable examples.

Ans:

| Supervised Learning                                                  | Unsupervised Learning                                     |
|----------------------------------------------------------------------|-----------------------------------------------------------|
| → It uses labeled data.                                              | → It uses unlabeled data.                                 |
| → The output is already known during training.                       | → The output is not known during training.                |
| → The model learns input-output learning.                            | → The model finds hidden patterns or groups.              |
| → Example: Predicting house price from area, location, no. of rooms. | → Example: Grouping customers based on buying behaviours. |

c) Why is data important in Machine Learning?

Ans: Data is important in Machine Learning because ml models learn patterns from data. If the data is accurate, relevant, and sufficient, the model can make better predictions. Poor-quality data can lead to poor results, even if the algorithm is good.

## Question 2: Measurement Scales and Attributes!

Ans: Given:

IQ score changed from 80 to 160.

Funding changed from \$50,000 to \$100,000

Execution time changed from 2 hours to 4 hours.

a) The researcher became twice as intelligent.

Ans: NOT CORRECT.

Because IQ is measured on an interval scale. IQ does not have a true zero. An IQ of 0 does not mean complete absence of intelligence.

b) The annual funding doubled.

Ans: CORRECT

Funding increased from \$50,000 to \$100,000. It has true zero point. So, the funding doubled.

c) The algorithm required twice as much time execution time.

Ans: CORRECT

Execution time increased from 2 hours to 4 hours, So, the algo required twice as much time.

d) The increase in IQ was 80 points.

Ans: CORRECT.

$$\text{Here, } 160 - 80 = 80$$

↑

Increment

So, the IQ score increased by 80 points.

e) Based on your reasoning, classify the variables (IQ score, annual funding, and execution time) as either interval or ratio scales.

Ans:

IQ score → Interval Scale (Differences are meaningful, but ratio comparison is not meaningful.)

Annual funding → Ratio score (It has true zero point)

Execution time → Ratio scale (It has true zero point)

f) Explain the significance of a true zero in determining whether ratio comparisons are meaningful.

Ans:

A true zero means complete absence of the quantity to be measured. For eg. \$0 means no funding, 0 hours means no execution time. Because funding and time have true zero values, ratio comparisons like "twice as much" are meaningful.

### Question 3: Data Representation

Consider the dataset  $X = \begin{bmatrix} [2, 3], [4, 1], [1, 8] \end{bmatrix}$   
 $\hat{S}, 6]$

a) Determine the dimensions of the dataset by specifying the number of samples and features.

Ans:-

Here,  $X = \begin{bmatrix} [2, 3], [4, 1], [1, 8] \end{bmatrix}$   
 $\hat{S}, 6]$ ,  
Samples = 4  
Features = 2

$\therefore$  dimension of the dataset =  $4 \times 2$

b) Represent the third sample as a column vector.

Ans:-

Third sample is  $(\hat{S}, 6]$   
As a column vector  $\begin{bmatrix} 5 \\ 6 \end{bmatrix}$

c) Identify the sample having the largest value in the second feature and specify its coordinates.

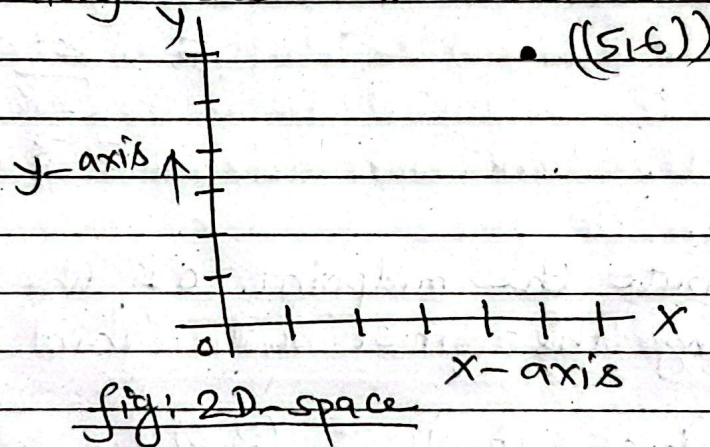
Ans: Here,

second features are:  $[3, 1, 6, 8]$   
The largest value is  $[8]$   
So, the sample is  $[1, 8]$ .

d) Explain the geometric interpretation of the point  $(5,6)$  in a two-dimensional feature space.

Ans:

The point  $(5,6)$  represents a data sample in 2D space. 5 is the 1st value and 6 is the 2nd value or feature. Geometrically, it is a point located at 5 unit along the x-axis and 6 units along the y-axis.



#### Question 4: Vector Operations and Distance Computation

Using the dataset above, let  $x_1 = (2,3)$  and  $x_2 = (5,6)$

Q) Compute  $x_2 - x_1$

Ans:

$$\begin{aligned} x_2 - x_1 &= (5,6) - (2,3) \\ &= (5-2, 6-3) \\ &= (3,3) \end{aligned}$$

b) Determine the Euclidean distance between the first and third samples.

Ans: 1

first sample = (2, 3)

third sample = (5, 6)

$$\begin{aligned}\text{Euclidean distance} &= \sqrt{(5-2)^2 + (6-3)^2} \\ &= \sqrt{3^2 + 3^2} \\ &= \sqrt{18} \\ &= 3\sqrt{2} \text{ unit}\end{aligned}$$

c) Compute the midpoint of the line segment joining the first and third samples.

Ans:

first sample = (2, 3)

third sample = (5, 6)

$$\begin{aligned}\therefore \text{midpoint of the line segment joining the} \\ \text{1st and 3rd samples} &= \left( \frac{2+5}{2}, \frac{3+6}{2} \right) \\ &= \left( \frac{7}{2}, \frac{9}{2} \right) \\ &= (3.5, 4.5)\end{aligned}$$

d) Explain the significance of Euclidean distance in ML applications.

d) Ans:

Euclidean distance measures the straight-line distance between two data points. It is used in many machine learning algorithms, especially distance-based algorithms such as K-nearest Neighbours, clustering, and similarity-based models. A smaller Euclidean distance means two data points are more similar.

### Question 5: Euclidean Norm and Unit Vectors

a) Compute the Euclidean Norm of  $[-3, 4]$ ,  $[1, 2, 2]$  and  $[1, 1, 1, 1]$  and also determine their unit vector.

Ans:

$$V_1 = [-3, 4]$$

$$\begin{aligned}\text{Euclidean Norm } (\|V_1\|) &= \sqrt{(-3)^2 + 4^2} \\ &= \sqrt{9 + 16} \\ &= \sqrt{25} \\ &= 5\end{aligned}$$

$$\text{Unit vector: } \hat{V}_1 = \frac{V_1}{\|V_1\|} = \frac{[-3, 4]}{5} = \left[-\frac{3}{5}, \frac{4}{5}\right]$$

$$V_2 = [1, 2, 2]$$

$$\begin{aligned}\text{Euclidean norm } (\|V_2\|) &= \sqrt{1^2 + 2^2 + 2^2} \\ &= \sqrt{1 + 4 + 4} \\ &= \sqrt{9} \\ &= 3\end{aligned}$$

$$\text{Unit vector: } \hat{V}_2 = \frac{V_2}{\|V_2\|} = \frac{[1, 2, 2]}{3} = \left[\frac{1}{3}, \frac{2}{3}, \frac{2}{3}\right]$$

b) Explain the geometric meaning of a unit vector and its importance in ML.

Ans:

A unit vector is a vector whose magnitude is exactly 1. It represents direction only, not size.

In ML, unit vectors are important because they help normalize data. The range of the data is reduced to a certain range after normalization. Normalization makes comparison between vectors easier and fairer.

Question 6: Cosine Similarity

Given  $x = [1, 2, 3]$  and  $y = [2, 4, 6]$

a) Calculate the cosine similarity between  $x$  and  $y$ .

Ans:

Here,

$$x = [1, 2, 3], y = [2, 4, 6]$$

$$\cos \theta = \frac{x \cdot y}{\|x\| \|y\|}$$

$$x \cdot y = [1, 2, 3] \cdot [2, 4, 6]$$

$$= 1 \times 2 + 2 \times 4 + 3 \times 6$$

$$= 2 + 8 + 18$$

$$= 28$$

$$\begin{aligned}\|x\| &= \sqrt{1^2 + 2^2 + 3^2} \\ &= \sqrt{1 + 4 + 9} \\ &= \sqrt{14}\end{aligned}$$

$$\begin{aligned}\|y\| &= \sqrt{2^2 + 4^2 + 6^2} \\ &= \sqrt{4 + 16 + 36} \\ &= \sqrt{56}\end{aligned}$$

Now,

$$\begin{aligned}\cos \theta &= \frac{x \cdot y}{\|x\| \|y\|} \\ &= \frac{28}{\sqrt{14} \times \sqrt{56}} \\ &= \frac{28}{\sqrt{784}} \\ &= \frac{28}{28}\end{aligned}$$

$$\therefore \cos \theta = 1$$

b) Interpret the result and explain what it includes indicates about the relationship between the two vectors.

Ans: Interpretation of result.  
From above, calculation

The cosine similarity is 1.

This indicates that the vectors point in exactly the same direction. As  $\cos \theta = 1$ , means either the vectors are parallel or coincident on each other.

c) One application of cosine similarity is search engines. When we use search engine to search something, it uses cosine similarity to show output.